



A SECURE DIGITAL IDENTITY WALLET FOR SOUTH AFRICA



Your official identity,
ready when you are.

Store, share and verify official credentials on web and mobile, so an identity check takes seconds instead of paperwork.

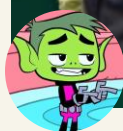
Final Demo

Tech Titans
COS 301 Capstone 2026



Meet the Team

Five minds. One Mission.



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Identity in South Africa **still runs on paper.**

Banks, hospitals, police, DLTCs and Home Affairs all depend on physical documents that can be lost, copied or faked.

THE GOAL

Proof of identity that is instant, private and impossible to fake, even with no signal.



Fragile documents

IDs and licences get lost, stolen or damaged, and replacing them takes weeks.



Copies everywhere

ID copies travel over WhatsApp and email with no control over where they end up.



Hard to trust

A photocopy cannot prove it is real, still valid, or that it belongs to the holder.



No signal, no check

Roadblocks, clinics and rural areas cannot count on a live connection.

One trusted digital credential. **Four simple steps.**



1

GOVERNMENT

**Issues a verified,
digitally signed
credential**



2

CITIZEN

**Stores it in the
FlashID wallet on a
trusted device**



3

CITIZEN

**Chooses exactly
which fields to
share**



4

ORGANISATION

**Scans the QR and
verifies it in seconds**



SECURE BY DESIGN



INSTANT
VERIFICATION



PRIVACY PROTECTED



BUILT FOR SA

Three audiences. Two apps. One source of truth.



Citizens

Mobile wallet

Web portal

- ✓ Wallet & selective-disclosure QR
- ✓ Emergency medical profile
- ✓ Security alerts & activity



Government admin

Web dashboard

- ✓ Register institutions & API keys
- ✓ Issue, revoke & reinstate
- ✓ POPIA audit log



Officials

Web portal

Mobile verifier

- ✓ Onboard citizens from the registry
- ✓ Scan & verify, online or offline
- ✓ Emergency break-glass access

BUILT WITH

Next.js · React Native (Expo) · .NET Clean Architecture · Azure SQL · Cosmos DB · Key Vault · Azure Face Liveness · Government Registry API

Locking the front door.



Logging in

Auto logout

5 wrong passwords lock the account for 30 minutes.

Password rules

10+ characters with upper, lower and a digit.

New device check

6-digit email code, valid 10 minutes, 5 tries.

BCrypt hashing

Work factor 12. We never store a password.

Same error message

"Invalid email or password", so emails aren't revealed.

Hashed secrets

OTPs and device tokens stored as SHA-256 hashes.



Sessions

HttpOnly cookie

JavaScript can't read the token

Logout everywhere

a token version bump kills every old session

Expiry

8 hours, or 30 days with remember me



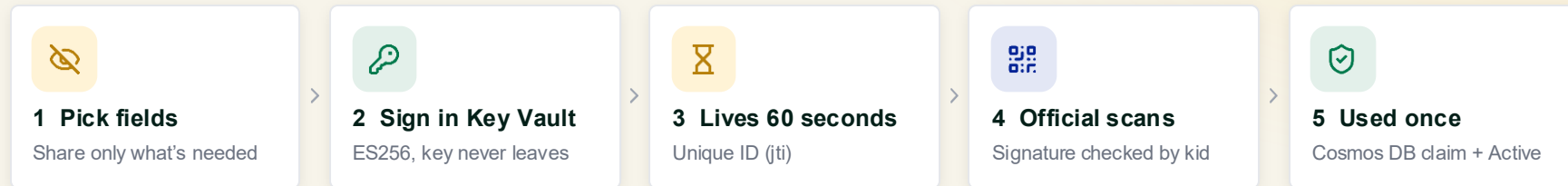
CSRF protection

Cookie `csrf_token` sent automatically

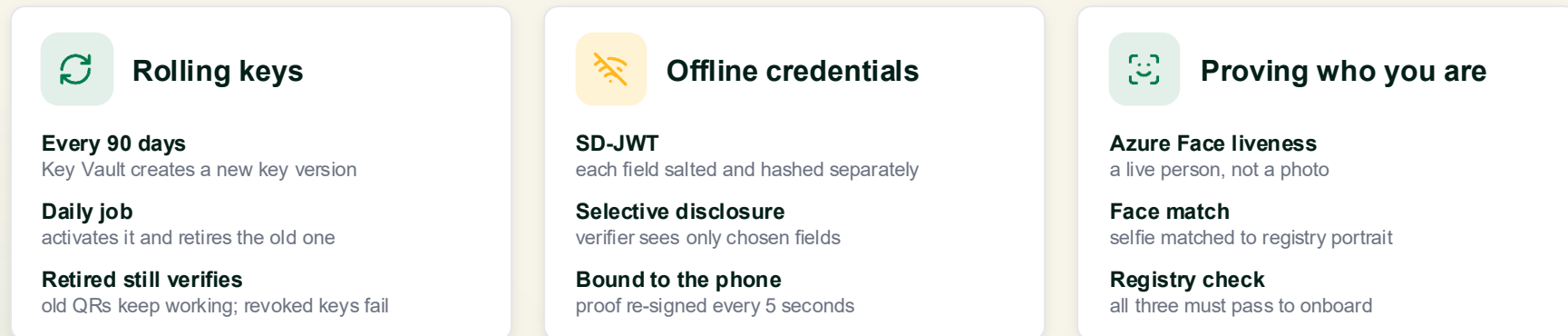
Header `X-CSRF-Token` only our page can set it

Must match, or the request gets 403.

A QR code that can't be faked, reused or copied.



If any check fails, the scan is rejected and logged. A screenshot or a second scan never works.



Watching, protecting and **keeping a trail.**



Stopping abuse

Rate limiting

10 logins a minute per IP and 300 a minute overall. Extra requests get 429.

Fraud scoring

Points for impossible travel, new devices and failed logins. High scores raise an alert.

Secure my account

Re-enter your password, then log out other devices or reset it.



Protecting data

SQL injection

EF Core parameters only. No raw SQL.

AES-256-GCM

Emergency fields encrypted at rest.

Certified PDFs

SHA-256 fingerprint catches any edit.

Mobile SecureStore

Keychain / Keystore, this device only.



Access & infrastructure

Role-based access

Citizen, official, government admin.

Registry API key

X-API-KEY, constant-time compare.

HTTPS + HSTS

Every request encrypted.

Safe errors

No stack traces in production.

Audit log

48 event types, reviewed by admins.

 FULL SYSTEM

LIVE DEMO

● FULL SYSTEM

GOVERNMENT ADMIN PORTAL

03

Wow factors

Three things a paper ID has never been able to do.



Offline QR

Verify with no signal on either phone.



Emergency QR

Medical info from a locked phone.



Impossible travel

Stops account takeover live.

Verified with zero bars, **on both phones.**

0

**internet connections needed
at the moment of verification**

SD-JWT · ES256 · device-bound

A copied or filmed code stops verifying
within about 90 seconds.



1 Prepared while online

The wallet receives a signed offline package and keeps it encrypted on the phone for up to 30 days.



2 Animated QR, bound to the phone

The phone's device key re-signs the code every 5 seconds, so a screenshot or recording is useless.



3 Verified on the official's phone

Issuer signature, revocation list and device binding are all checked locally.



4 Synced when back online

The result is written to the audit trail exactly once.



When you can't speak, **your phone can.**



One tap from the lock screen

A Quick Settings tile shows the code without unlocking the phone.



Signed by the phone's hardware

A StrongBox key signs each code. It is short-lived and single-use.



Officials only, break-glass

A reason is required. Only the medical fields and contacts the citizen chose are shown.



Everyone is told

The citizen and their emergency contacts are emailed, even after offline access.

Nobody flies Johannesburg to London in 22 minutes.



Johannesburg, ZA

Sign-in · 10:02

KNOWN



≈ 9,000 km in 22 minutes



London, UK

Sign-in · 10:24

SUSPICIOUS

≈ 24,500

km/h needed

900

km/h limit (faster than a commercial flight)



Instant security alert

The citizen sees the route on a map and why it was flagged.



QR generation frozen

A high-risk event blocks new QR codes for 30 minutes, so a thief cannot present the ID.



Secured in one step

After the citizen re-enters their password, all sessions end, the suspicious device is removed and the block lifts.



PROVE YOURSELF IN A FLASH.

Thank you. Questions?

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Tech Titans · flashid.co.za